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# Chapter 4

# Introduction to Cisco IOS

When preparing for the Cisco CCNA certification, you will be tested on a number of advanced topics, but before you get there it is important that you have a sound foundation. This chapter is designed to test you on the basics of the Cisco IOS and prepare you for related questions you will find on the ICND1 exam.

## The Problems You'll Work On

In this chapter, you'll review questions concerning the following topics:

- Recognizing the different port types found on a Cisco device and their purpose
- Understanding the different types of memory and what is stored there
- Navigating the Cisco IOS
- Grasping the basics of the Cisco device boot process
- Comprehending the information outputted from the `show version` command

## What to Watch Out For

Don't let common mistakes trip you up; watch for the following when working with these questions:

- Cisco devices will have different types of ports, but the most common are the console (CON) port and the auxiliary (AUX) port. The console port is used for local administration, whereas the auxiliary port is used to connect a modem to a router and to remotely manage the router.
- Serial ports are used to connect to a WAN environment by connecting to an external CSU/DSU. Serial ports can also be used to connect two routers directly together with a back-to-back serial cable. In this example, one router will act as the DCE device and the other will act as the DTE device (specified on each end of the cable). The DCE device will need to specify the clock rate command.
- There are different types of memory on the Cisco device. Flash memory is used to store the Cisco IOS; NVRAM is used to store the startup configuration; RAM or VRAM is used to store the running configuration; and ROM is used to store the POST routines, bootstrap program, and mini-IOSes, such as ROMMON, which you can use to troubleshoot problems with the real IOS.

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## Device Ports

***124–138*** Express the given number in scientific notation.

**124.** Which of the following represents a purpose for the serial port on a Cisco router?

(A) To connect to the LAN

(B) To connect the console port

(C) To connect to the WAN

(D) To connect a modem

**125.** Which of the following port identifiers are used to reference a Fast Ethernet port? (Select two.)

(A) s0/0

(B) f0/0

(C) con 0

(D) aux 0

(E) fa0/0

**126.** What type of cable is used to connect to the console port?

(A) Crossover

(B) Straight-through

(C) Null modem

(D) Rollover

**127.** Your manager notices an AUX port on the back of the Cisco router and asks what it is used for. How would you reply?

(A) To connect a CSU/DSU

(B) To connect an external Ethernet interface

(C) To connect an internal Ethernet interface

(D) To connect a modem

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**128.** You are performing a security audit on the network devices and have verified that the console port has a password configured. What other port should you verify has a password?

(A) FastEthernet0/0

(B) FastEthernet0/1

(C) AUX

(D) Serial0/0/0

**129.** You are configuring the DCE end of a point-to-point serial link between two routers. What extra command is typed on the DCE end of the link?

(A) `clock rate 64000`

(B) `encapsulation hdlc`

(C) `ip address w.x.y.z`

(D) `no shutdown`

**130.** The external CSU/DSU connects to which port on the router?

(A) FastEthernet0/0

(B) Con 0

(C) Aux 0

(D) Serial0/0/0

**131.** What type of module can be purchased to install an internal CSU/DSU into the router?

(A) WNA

(B) WCI

(C) WIC

(D) WAC

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**132.** You are trying to assign an IP address to the LAN port on the router and get the following message on the screen.

```
Router(config)#interface e0
^
% Invalid input detected at '^' marker.
```

You use the show version command to try to identify the problem. The output is displayed below. What is the problem?

```
Cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes
of memory
Processor board ID JAD05190MTZ (4292891495)
M860 processor: part number 0, mask 49
2 FastEthernet/IEEE 802.3 interface(s)
4 Low-speed serial(sync/async) network interface(s)
239K bytes of NVRAM
62720K bytes of processor board System flash (Read/Write)
Configuration register is 0x2102
Router>
```

This means:

(A) The interface is disabled.

(B) The interface type specified does not exist.

(C) You are at the wrong prompt.

(D) The interface is already configured.

## Memory Types

***133–138*** Express the given number in scientific notation.

**133.** Where is the running configuration stored?

(A) NVRAM

(B) VRAM

(C) ROM

(D) Flash

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**134.** Which of the following are stored in ROM on the Cisco device?

(A) Startup config

(B) POST

(C) Bootstrap

(D) IOS

(E) ROMMON

(F) Running config

**135.** What type of memory is shown here?

(A) VRAM

(B) NVRAM

(C) ROM

(D) Flash

**136.** Which commands can you use to save the running configuration to the startup configuration? (Select two.)

(A) `copy startup-config running-config`

(B) `save running-config startup-config`

(C) `save running-config`

(D) `copy running-config startup-config`

(E) `write`

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**137.** When the Cisco router starts up, where does it find the Cisco IOS?

(A) NVRAM

(B) RAM

(C) ROM

(D) Flash

**138.** When you first boot up the Cisco router, where does it locate the startup configuration?

(A) NVRAM

(B) RAM

(C) ROM

(D) Flash

## IOS Basics

***139–150*** Express the given number in scientific notation.

**139.** You are connected to your switch and wish to remotely administer the Cisco router. Which command would you use?

(A) `telnet`

(B) `enable`

(C) `connect`

(D) `rdp`

**140.** You are at the global configuration prompt and wish to configure a communication protocol for remote access to the router. What command would you type?

(A) `line vty 0 4`

(B) `line ssh 0`

(C) `line telnet 0 4`

(D) `line telnet 0`

**141.** What command would you use to navigate from priv exec mode to user exec mode?

(A) `disable`

(B) `shutdown`

(C) `enable`

(D) `exit`

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**142.** You wish to get help on which parameters exist on the ping command. How would you get a list of parameters?

(A) `help ping`

(B) `ping ?`

(C) `? ping`

(D) `?? ping`

**143.** What is the command to move to priv exec mode from user exec?

(A) router`>enable`

(B) router#`enable`

(C) router(config)#`enable`

(D) router#`disable`

**144.** Which of the following prompts represents global configuration mode?

(A) router#

(B) router(config)#

(C) router>

(D) router(global)#

**145.** What is the default router name if a name is not assigned to the router during initial configuration?

(A) R1

(B) switch

(C) router

(D) default

**146.** What command would you use to navigate to global configuration mode?

(A) `>config terminal`

(B) `#setup`

(C) `>enable`

(D) `#config terminal`

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**147.** What types of actions can an administrator perform at this IOS prompt? (Select two.)

```
R1(config-if)#
```

(A) `Show running-config`
(B) Disable the interface
(C) Show the IOS version
(D) Change the hostname
(E) Assign an IP address

**148.** You wish to run the initial configuration dialog. What command would you use?

(A) install
(B) dialog
(C) setup
(D) initdialog

**149.** Under what conditions would the initial configuration dialog appear?

(A) When there is no running-config
(B) When there is no startup-config
(C) When you type initdialog
(D) When you type setup

**150.** You have connected a computer to the console port of a brand new router. What will the router display on the screen after it is powered on?

(A) router>
(B) POST errors
(C) The option to enter initial configuration
(D) router#

## Boot Process

***151–156*** Express the given number in scientific notation.

**151.** What is the purpose of the POST?

(A) Locate the IOS in flash memory.
(B) Locate the IOS in ROM.
(C) Verify that hardware is functioning.
(D) Apply the startup configuration.

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**152.** You have rebooted your Cisco switch and it displays an amber light on the system LED during POST. What does this indicate?

(A) A failure during POST

(B) System booted without fault

(C) A successful POST

(D) Failed communication with router

**153.** What happens when a router starts up and it does not have a startup configuration file?

(A) The router prompts for a password.

(B) The router prompts to enter setup mode.

(C) The router applies a default configuration.

(D) The router goes into global configuration mode.

**154.** Which of the following best describes ROMMON?

(A) A limited operating system used to troubleshoot startup and password recovery

(B) Storage area to hold the startup configuration

(C) Used to store the Cisco IOS

(D) Storage area to hold the running configuration

**155.** Which of the following best identifies the boot order of a Cisco device?

(A) Bootstrap loads IOS from flash; POST; apply startup configuration to running config.

(B) Bootstrap loads IOS from flash; apply startup configuration to running config; POST.

(C) POST; apply startup configuration to running config; bootstrap loads IOS from flash.

(D) POST; bootstrap loads IOS from flash; apply startup configuration to running config.

**156.** What keystroke is used to access the ROMMON prompt in order to bypass the password?

(A) Ctrl-Alt-Del

(B) Ctrl-Del

(C) Ctrl-Enter

(D) Ctrl-Break

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## IOS Version Information

***157–161*** Express the given number in scientific notation.

***157.*** Using this output of the show version command, how many Fast Ethernet ports exist on the router?

```
Cisco 2811 (MPC860) processor
(revision 0x200) with 60416K/5120K bytes of memory

Processor board ID JAD05190MTZ (4292891495)

M860 processor: part number 0, mask 49

2 FastEthernet/IEEE 802.3 interface(s)

4 Low-speed serial(sync/async) network interface(s)

239K bytes of NVRAM

62720K bytes of processor board System flash (Read/Write)

Configuration register is 0x2102

Router>
```

(A) 49

(B) 2

(C) 1

(D) 4

***158.*** Using the output below, what version of the Cisco IOS is running?

```
Router>show version

Cisco IOS Software, 2800 Software (C2800NM-ADVIPSERVICESK9-M), Version
12.4(15)T1, RELEASE SOFTWARE (fc2)

Technical Support:http://www.cisco.com/techsupport

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```

(A) 2800

(B) 12.4

(C) 15

(D) T1

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**159.** Your manager is wondering how many Fast Ethernet ports are on the router. What command would you use to find that out?

(A) `show help`

(B) `show modules`

(C) `show fastethernet`

(D) `show version`

**160.** You wish to verify the version of the Cisco IOS you are running. What command would you use?

(A) `Show IOS`

(B) `Show memory`

(C) `Show version`

(D) `Show software`

**161.** You wish to view the current value of the configuration register. What command would you use?

(A) `show reg`

(B) `show version`

(C) `show flash`

(D) `show config`